

Composing Functions from Symbolic Rules

Answer Key

Algebra 2

Quick Answers

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|--|--|
| 1. (a) $g(4) = 13$, (b) $f(g(4)) = 31$ | 8. (a) the composed rule, expanded = |
| 2. $3x + 11$ | $x^3 + 3x^2 + 3x + 1$, (b) the other order at $x = 2$ |
| 3. (a) the other order = $3x + 3$, — | = 9, — |
| 4. (a) $f(g(2)) = 16$, (b) $g(f(2)) = -2$ | 9. 4, -10 |
| 5. $4x^2 - 12x + 10$ | 10. (a) the composed rule = $\sqrt{x^2 + 4}$, (b) its |
| 6. 7 | value at $x = 3 = 3.61$, — |
| 7. $\frac{2}{x+2}$ | |
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What is verified

14 of 17 answers machine-checked against SymPy, across 10 problems.

— marks an answer only you can judge — problems 3, 8, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 2

HSF-BF.A.1c – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Difficulty 1–5 of 5 across 17 tagged checks.

1. $f(x) = 2x + 5$, $g(x) = x^2 - 3$. (a) $g(4)$. (b) $f(g(4))$.

Solution (a): Run the inner machine first: $g(4) = 4^2 - 3 = 16 - 3$.

$g(4) = 13$

Solution (b): Feed that output into f : $f(13) = 2(13) + 5 = 26 + 5$.

$f(g(4)) = 31$

2. $f(x) = 3x - 1$, $g(x) = x + 4$. Find $(f \circ g)(x)$.

Solution: Substitute the whole rule for g into f 's formula, keeping it bracketed:

$$f(g(x)) = 3(x + 4) - 1 = 3x + 12 - 1.$$

$(f \circ g)(x) = 3x + 11$

3. Same f and g . (a) $(g \circ f)(x)$. (b) Do the orders agree?

Solution (a): Now f runs first, and its output goes into g :

$$g(f(x)) = (3x - 1) + 4 = 3x + 3.$$

$$(g \circ f)(x) = 3x + 3$$

Solution (b): *Instructor-judged.* No: $3x + 11$ and $3x + 3$ are different rules, so $f \circ g$ and $g \circ f$ are different functions. The cause is which operation happens first. In $f \circ g$ the $+4$ happens before the multiplication by 3, so the 4 gets tripled and contributes 12; in $g \circ f$ it is added after the tripling and contributes only 4. Full credit needs both the comparison and that reason.

4. $f(x) = x^2$, $g(x) = x - 6$. (a) $(f \circ g)(2)$. (b) $(g \circ f)(2)$.

Solution (a): g first: $g(2) = 2 - 6 = -4$; then $f(-4) = (-4)^2$.

$$16$$

Solution (b): f first: $f(2) = 4$; then $g(4) = 4 - 6$.

$$-2$$

5. $f(x) = x^2 + 1$, $g(x) = 2x - 3$. Find $(f \circ g)(x)$ expanded.

Solution: Substitute, then expand the square carefully:

$$\begin{aligned} f(g(x)) &= (2x - 3)^2 + 1 \\ &= 4x^2 - 12x + 9 + 1. \end{aligned}$$

$$4x^2 - 12x + 10$$

6. $f(x) = 2x + 5$, $g(x) = x - 1$. Solve $(f \circ g)(x) = 17$.

Solution: Build the composed rule first, then solve the ordinary equation:

$$f(g(x)) = 2(x - 1) + 5 = 2x + 3, \quad 2x + 3 = 17.$$

So $2x = 14$. Check: $g(7) = 6$ and $f(6) = 17$ ✓

$$x = 7$$

7. $f(x) = \frac{x}{x+1}$, $g(x) = \frac{2}{x}$. Find $(f \circ g)(x)$.

Solution: Substitute $\frac{2}{x}$ for every x in f 's rule, then clear the compound fraction by multiplying top and bottom by x :

$$f(g(x)) = \frac{\frac{2}{x}}{\frac{2}{x} + 1} = \frac{\frac{2}{x} \cdot x}{\left(\frac{2}{x} + 1\right) x} = \frac{2}{2 + x}.$$

$$\frac{2}{x+2}$$

8. $f(x) = x^3$, $g(x) = x + 1$. (a) $(f \circ g)(x)$ expanded. (b) $(g \circ f)(2)$. (c) Why order changes the function.

Solution (a): $f(g(x)) = (x + 1)^3$, and expanding gives

$$x^3 + 3x^2 + 3x + 1$$

Solution (b): f first: $f(2) = 8$; then $g(8) = 8 + 1$.

$$9$$

Solution (c): *Instructor-judged.* At $x = 2$ part (a) gives $(2 + 1)^3 = 27$, while part (b) gives 9. In $f \circ g$ the 1 is added *before* cubing, so it is inside the cube; in $g \circ f$ it is added *after*, so it is merely added on. The rules themselves differ — $x^3 + 3x^2 + 3x + 1$ against $x^3 + 1$ — so the disagreement is not a coincidence at one input: the two orders build different functions. Full credit identifies which operation runs first in each order.

9. $f(x) = x^2$, $g(x) = x + 3$. Solve $(f \circ g)(x) = 49$.

Solution: The composed rule is $(x + 3)^2$, so $(x + 3)^2 = 49$ and $x + 3 = \pm 7$. Both signs give a genuine solution, since squaring accepts either.

$$x = 4 \text{ or } x = -10$$

10. Challenge: $f(x) = \sqrt{x - 1}$, $g(x) = x^2 + 5$. (a) $(f \circ g)(x)$. (b) Its value at $x = 3$. (c) Why every real input works.

Solution (a): Substitute the inner rule under the radical:

$$f(g(x)) = \sqrt{(x^2 + 5) - 1} = \sqrt{x^2 + 4}.$$

$$\sqrt{x^2 + 4}$$

Solution (b): $\sqrt{3^2 + 4} = \sqrt{13} = 3.60555\dots$

$$3.61$$

Solution (c): *Instructor-judged.* On its own f needs $x - 1 \geq 0$, so it fails for every input below 1. But the composition never hands f such an input: g outputs $x^2 + 5$, which is at least 5 for every real x , comfortably above 1. After the subtraction the radicand is $x^2 + 4$, which is at least 4 and so never negative. Full credit connects the inner function's range to the outer function's domain requirement.